

Full Stack Developer Assessment

This assessment is designed to evaluate a candidate's Full Stack development skills through the design and development of a flight comparison prototype. The candidate is expected to demonstrate end-to-end development capabilities, integration with multiple data sources, data aggregation and comparison, user experience, problem-solving, testing, and error handling. Actual provider integration is preferred to the extent practically possible, and candidates are encouraged to explore different legitimate integration options. AI may be leveraged as needed to accelerate development, research, and problem-solving, while the candidate remains responsible for and should fully understand the final solution.

Flight Comparison Application

1. **Objective:** Develop a prototype application that allows users to search and compare flight options from:
 - MakeMyTrip
 - Cleartrip
 - Goibibo
 - IndiGo
 - Air India Express
2. **Search Criteria:** The application should accept:
 - Source
 - Destination
 - Travel date
 - Preferred time range
3. **Expected Results:** For each flight option, display:
 - Provider
 - Airline
 - Flight number
 - Departure and arrival time
 - Flight duration
 - Number of stops
 - Price
 - Benefits or offers

Users should be able to sort, filter, and compare results based on price, flight time, departure time, benefits, and overall value.

4. **Provider Integration:** Actual integration with the specified providers is preferred as much as practically possible. Candidates are encouraged to explore different legitimate integration options for obtaining current flight information. Where actual integration is not possible due to access, authentication,

commercial, technical, or other external limitations, an appropriate alternative or representative data may be used. The candidate should document:

- Providers successfully integrated
- Integration option used for each provider
- Providers not integrated and the reason
- Any alternative or representative data used

5. **AI Usage:** Candidates are encouraged to leverage AI as needed throughout the assignment. AI may be used to accelerate development, research integration options, troubleshoot issues, improve the application, generate or review tests, and assist with documentation. The candidate remains responsible for the final solution and should be able to:

- Explain the solution
- Explain important implementation decisions
- Demonstrate and troubleshoot the application
- Identify limitations and areas for improvement
- Describe how AI was used during development

Effective use of AI will be considered part of the candidate's overall problem-solving and development capability.

6. **Basic Functionality:** The application should:

- Search based on the entered criteria
- Aggregate results from multiple providers
- Present results consistently
- Allow sorting and filtering
- Compare price, flight time, and benefits
- Handle the same flight available through multiple providers
- Handle invalid inputs
- Handle unavailable providers
- Handle partial results
- Handle cases where no flights are found

7. **Deliverables**

- Complete source code
- Working prototype
- Instructions to run the application
- Relevant tests
- Brief documentation covering integrations, assumptions, limitations, and AI usage

8. **Evaluation:** The assessment will consider:

- Functionality and user experience
- Actual provider integration
- Ability to work with different integration options
- Aggregation and comparison of data
- Problem-solving
- Effective use of AI
- Quality and organization of the solution
- Error handling
- Testing
- Documentation

Candidates should be prepared to demonstrate the application, explain their integration choices and key implementation decisions, discuss challenges encountered, and describe how AI helped them develop the solution.